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Hybrid Black-Box Classification for Customer Churn Prediction with Segmented Interpretability Analysis

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Abstract

Customer retention management relies on advanced analytics for decision making. Decision makers in this area require methods that are capable of accurately predicting which customers are likely to churn and that allow to discover drivers of customer churn. As a result, customer churn prediction models are frequently evaluated based on both their predictive performance and their capacity to extract meaningful insights from the models. In this paper, we extend hybrid segmented models for customer churn prediction by incorporating powerful models that can capture non-linearities. To ensure the interpretability of such segmented hybrid models, we introduce a novel model-agnostic approach that extends SHAP. We extensively benchmark the proposed methods on 14 customer churn datasets on their predictive performance. The interpretability aspect of the new model-agnostic approach for interpreting hybrid segmented models is illustrated using a case study. Our contributions to decision making literature are threefold. First, we introduce new hybrid segmented models as powerful tools for decision makers to boost predictive performance. Second, we provide insights in the relative predictive performance by an extensive benchmarking study that compares the new hybrid segmented methods with their base models and existing hybrid models. Third, we propose a model-agnostic tool for segmented hybrid models that provide decision makers with a tool to gain insights for any hybrid segmented model and illustrate it on a case study. Although we focus on customer retention management in this study, this paper is also relevant for decision makers that rely on predictive modeling for other tasks.

Keywords: XAI, customer retention, hybrid algorithm, explainable analytics

1. Introduction

A customer base is the most important asset for marketing decision makers. With their decisions, they aim to generate value by attracting more customers (i.e., customer acquisition), generating more value from existing customers (i.e., customer development) or by retaining more customers (i.e., customer retention). In this paper we focus on customer retention, as extensive research has demonstrated the pivotal significance of customer retention efforts. Retaining customers has been found to be considerably more cost-effective compared to attracting new customers [1]. Furthermore, churners can also have a negative impact on other customers within their network, potentially exacerbating churn further [2].

To establish effective customer retention campaigns, the initial steps involve identifying the individuals prone to churn (i.e., 'who' is at risk?) and understanding the underlying reasons behind their potential churn (i.e., 'why' at risk?) [3]. Firms can rely on advanced analytics to answer these questions and in the literature this is known as customer churn prediction modeling [e.g., 4]. Consequently, customer churn prediction models have two key requirements. First, they need to accurately identify customers who are at risk. This aspect is evaluated based on predictive performance measures such as accuracy, the area under the receiver operating characteristic (ROC) curve (AUC), or managerial relevant metrics like top decile lift (TDL) or Expected Maximum Profit Classification Performance Measure (EMPC) [5]. Secondly, these models should provide insights into the reasons behind customer churn, which is linked to the model interpretability.

Although certain models, such as decision trees or logistic regression, inherently offer interpretability, many other methods are considered "black-box" models, including most ensemble methods (e.g., Random Forests) and neural network-based approaches. Despite their higher performance on complex non-linear tasks, black-box models' limited interpretability has been a significant factor hindering their adoption. Nevertheless, with the increasing complexity of customer data mined from historical tabular repositories [6], text [7], or even graphs [8], black-box models are becoming increasingly common in standard customer analytics toolboxes. An intriguing position within this dichotomy of black-box versus white-box models is occupied by so-called *hybrid models* ,which are also known as two-stage models [4]. Hybrid models refer to the combination of disparate modeling techniques to solve a particular problem. The resulting hybrid models leverage the strengths of their member models to improve accuracy, robustness, and interpretability.

To flexibly gain insight from any black-box modeling component, the utilization of model-agnostic interpretability techniques such as feature importance scores, partial dependence plots, or Shapley Additive Explanations (SHAP) values [9] is necessary. While previous hybrid models that combine white-box techniques have achieved remarkable results in churn prediction [4], no previous studies have attempted to integrate and explain powerful black-box components into the hybrid paradigm. We believe this extension to be a logical next step accommodating the increasingly complex featurization of customer analytics.

In this study, we bridge this gap, and extend hybrid segmented modeling with black box components for customer churn prediction. We ensure interpretability by proposing a model agnostic approach to derive insights from hybrid models, even if those models are inherently not interpretable. Specifically, we propose an extension of the popular Logit Leaf Model (LLM) [4] with Random Forests (RF), Extreme Gradient Boosting (XGB), Artificial Neural Networks (ANN), Tabnet and Catboost as black-box components, and propose a new method to derive insights from hybrid segmented models by extending the model-agnostic interpretability SHAP method [9]. We extensively benchmark the proposed methods on a 14 customer churn data sets and evaluate predictive performance using Area Under the Curve (AUC) and Top Decile Lift (TDL). Finally, the interpretability aspect of the new model-agnostic approach for segmented models is illustrated using a case study.

Our main contributions are threefold. First, we extend hybrid segmented modeling in customer churn prediction literature by proposing five hybrid segmented methods that rely on blackbox models (i.e., RF, XGB, ANN, Tabnet or Catboost), and benchmark their performance with the respective base models on our set of fourteen customer churn data sets. Second, the performance of the newly proposed hybrid models is compared with the LLM. As more complex models have achieved important gains in predictive performance and become more widespread as a result of the necessity to include complex data sources in predictive models, it is important to explore complex models for segmented hybrid models as well. Third, we propose a model-agnostic approach to derive interpretable output for decision makers from hybrid segmented models based on SHAP. The aspect of interpretability is exemplified through a comprehensive case study.

The structure of this paper is as follows. In the next section, we review the relevant literature on customer churn prediction and hybrid segmented modeling. Section 3 introduces interpretable extended segmented modelling as a novel hybrid interpretable methodology for churn prediction. In section 4 we outline the design of the study to empirically benchmark the hybrid segmented models against their base classifiers and the LLM. The results of these analyses and an illustrative case study are discussed in Section 5. Section 6 concludes the paper and presents areas for future research.

2. Related Literature

In this discussion of related literature, we subsequently discuss (i) classification methods deployed in CPP, followed by an in-depth discussion of the recently emerging (ii) hybrid segmented classifiers, and finally, (iii) explainability in CCP.

2.1. Classifiers for Customer Churn Prediction

Customer churn prediction has received considerable attention in multiple literature streams over the past decades. Many contributions have sought to improve the predictive performance of such models, inspired by the belief that small changes in predictive accuracy are often observed to be *managerially meaningful* [10]. The literature has shown that many, if not all, steps involved in the development of a CCP model can potentially affect predictive performance, including data preprocessing [11], data selection, feature selection [12], feature engineering [13] and model evaluation [5]. However, the literature has been dominated by work investigating the impact of the methodology deployed to deliver predictive CCP models. Since CCP is generally modelled as a binary classification problem (will the customer churn or not), much work is set out to introduce, modify, and compare new classifiers [e.g., 4, 14].

It is generally understood that when CCP is adopted to support decision-making, models should ideally reconcile predictive accuracy and interpretability, an important subdimension of explainability. Classifiers that feature both qualities have thus become natural choices in literature and practice. This includes primarily (i) statistical regression methods such as logistic regression and regularized counterparts, such as LASSO regression [15], and (ii) decision trees such as CART and C4.5, but also other methods such as (iii) naive Bayes, and (iv) k-nearest neighbors. Nevertheless, drawn by the promise of enhanced accuracy, machine learning has been widely solicited as a supplier of performance black box classifiers. A first noteworthy category are *ensemble* learners, known to merge several *weak* member classifiers into an overarching, more flexible yet stable predictive model. *Homogeneous* ensemble learners such as Random Forest [16], Rotation Forest [17] and XGB [18], which in casu combine decision tree-like base models, have been shown to be particularly competitive for CCP and are generally still deemed to be among the most potent candidate learning algorithms [18]. More recently, heterogeneous ensemble learners that combine various base algorithms and often deploy optimization to inspire a selective or weighted member combination emerged in the CCP literature [19]. A second notable category is deep learning, proven to be especially capable of accommodating non-tabular [13] and unstructured [7] data and to substitute often heuristic feature engineering required to process such data in conventional classifiers. Recently, specialized attention-based [20] neural networks were introduced in an effort to increase the effectiveness of neural networks in learning from tabular data [21]. Among these innovative architectures, Tabnet [22] has demonstrated remarkable efficacy [23, 24]. However, there are no established benchmarks that evaluate these methods for predicting churn.

2.2. Hybrid Segmented Modeling

Table 1: Literature review: applications of hybrid segmented learners in CCP

				Method characteristics				
Reference	**Method**	**Method components**	**Interpretation method**	**Hybrid**	**Segmented**	**Supervised**	**BB**	**XAI**
[25]	Clusterwise boosted C5.0	(1) Unsupervised learning cluster methods, (2) C5.0 trees with boosting	None	Yes	Yes	No	Yes	No
[26]	Clusterwise rule induction	(1) K-means clustering; (2) FOIL inductive rule learning	Inherent[2]	Yes	Yes	No	No	No[2]
[4]	Logit leaf model (LLM)	(1) C4.5; (2) Logistic regression with stepwise feature selection	Inherent	Yes	Yes	Yes	No	Yes
[27]	Clusterwise classification	(1) Rough K-means; (2) various classifiers (naive bayes, random forest, k-nearest neighbors, support vector machines and decision trees)	None	Yes	Yes	No	Yes	No
[28]	Clusterwise neural networks	(1) Proposed probabilistic possibilistic fuzzy C-means; (2) Levenberg–Marquardt-based ANN	None	Yes	Yes	No	Yes	No
[29]	Uplift LLM	(1) Uplift decision tree; (2) Uplift logit	Inherent	Yes	Yes	Yes	No	Yes
[30]	Spline-rule ensembles with sparse group lasso	(1) Penalized cubic regression splines; (2) CART; (3) sparse group lasso	Inherent	Yes	No[3]	Yes	No	Yes
This study	Hybrid black box classifiers	(1) C4.5; (2) Random Forest, XGB, MLP, Tabnet	Post-hoc	Yes	Yes	Yes	Yes	Yes

Notes: This literature review only considers journal publications discussing CCP applications that involve customer segmentation in a first stage, followed by segment-wise classification. Method classification dimensions: *Hybrid* = the classifier deploys a two-stage method; *Segmented* = the final model takes the form of a clusterwise classifier; *Supervised* = the hybrid classifier is fully supervised; *BB* = black box method; *XAI* = the hybrid learner supports interpretability. The (index) in the *Method* column denotes the model training stage. When multiple methods are specified within a stage, the reference paper proposed alternative options. [1] The study features an analysis of churn indicator importance which is not based on the hybrid model, but rather on an independent logistic regression model.

[2] The method's potential for interpretability was not demonstrated in this study.

[3] Cluster membership indicators are used as features in the 2^{nd}-stage classifier.

A recently emerging subclass of supervised learners in churn prediction literature are hybrid segmented models. Such models share a capacity to naturally accommodate the existence of clusters of customers who churn for different reasons, or who simply share important characteristics. Methods in this class typically involve two stages: a segmentation stage in which customers are assigned to homogeneous subgroups, followed by a segments-specific classifier training stage.

Table 1 provides an overview of applications of hybrid segmented models in customer churn prediction. The methods they feature can be classified in three groups according to the segmentation strategy chosen. First, several applications rely on unsupervised learning (cluster analysis) as a segmentation strategy. Bose and Chen [25] compare 5 cluster methods, including including K-means, K-medoids, self-organizing maps, fuzzy c-means and BIRCH, and deploy boosted C5.0 trees to train segmentwise classifiers. K-means is also deployed in [26] and [27] while [28] opts for fuzzy c-means. A second group relies on supervised learning. In this category, we find the LLM that has seen multiple applications in CCP [4, 29]. A final group are approaches that use cluster membership indicators as new features in a classifier. This category includes spline-rule ensembles found in [30] as well as a model variant discussed in [25].

From the overview in Table 1 it is clear that black box classifiers have never been deployed in hybrid segmented learners that pursue interpretation and explainability. The three methods in this overview that relied on black box classifiers focused solely on predictive performance. The framework we propose in this paper intends to reconcile the concept of interpretable hybrid segmented learning with powerful black box classifiers by taking advantage of the recent surge in versatile and powerful post-hoc model interpretation techniques. In particular, we build upon SHAP, a powerful method for post-hoc model interpretation.

Our framework extends the analytical toolkit of decision makers. First, the first stage of the hybrid model that derives the segments remains completely interpretable as this is derived from a decision tree. This would not be the case if one creates a full black-box model. Hence, decision makers could thus still interpret the different segments, reflecting a relevant decision level for customer retention efforts. Second, the segment-specific models of the hybrid models return the final prediction for each customer, for which optimal performance may be obtained using black box models. In the proposed framework, we allow for model agnostic interpretation of the segment specific models based on SHAP. This is relevant for decision makers as they can plug in any method for a given segment, and are not constrained to logistic regression as in the logit leaf model.

2.3. Explainability in Customer Churn Prediction Methods

In churn prediction, understanding the decisions of the models is of paramount importance for decision effectiveness. Classic CCP models such as decision trees and linear regression are inherently interpretable, exhibiting properties such as linearity and monotonicity. On the other hand, the internal workings of neural networks and ensemble models are not easily understood. In this case, explainability methods can be used to 'open the black box' and provide informative explanations. Explainability is rapidly becoming a cornerstone in the design of churn prediction models [31]. Some explanation mechanisms are model-specific (e.g., attention [22]), others are model-agnostic (e.g., SHAP). Model-agnostic explanations are generated after model training (post hoc), and are applicable to any predictive model. In problem domains where 'free lunches' [32] exist, i.e., one algorithmic class consistently outperforms all others, an algorithm-specific explainability method for the dominant modeling class suffices. For example, in transformer-based black-box hybrid models, one can consistently turn towards attention weights to explain the transformer component. Examples can be found in domains such as business [13], vision [33] or natural language processing [34]. Conversely, for churn prediction previous research has shown that there is no free lunch, i.e., the optimal algorithm varies widely across datasets and settings [35]. As such, for churn prediction model-agnostic methods are preferable. This is reflected in recent efforts in the literature; [36, 37] and [38] used Shapley values and SHAP to explain churn prediction models, respectively. To the best of our knowledge, and as depicted in Table 1, hybrid segmented modeling has so far only relied on inherent interpretability.

3. Method

In this section, we introduce segmented hybrid models for predicting customer churn. Next, we discuss how we extend these models and highlight the importance for a model-agnostic approach to derive interpretable insights out of such models.

3.1. Logit Leaf Model

Customer churn prediction is a domain that benefits from segmented, hybrid models as such segment specific insights are valuable for decision makers in this domain. If one can, for example, detect segments with different drivers, retention campaigns could be designed for each segment specifically. In a seminal paper in the domain, the LLM is introduced as an inherently interpretable hybrid segmented model [4]. It cleverly combines decision trees and logistic regressions such that the final model returns a decision tree that can be considered as a segmentation, and at each leaf node a segment specific logistic regression allows to gain insights in segment specific drivers of churn. As both building blocks are inherently interpretable, the resulting model is also inherently interpretable.

The LLM relies on supervised learning for both defining the segments and constructing segment-specific models. By an internal cross-validation mechanism, a decision tree is pruned followed by segment-specific logisitic regressions. De Caigny et al. [4] show that the LLM performs significantly better than both its building blocks, decision trees, and logisitic regression. This might be explained by the fact that decision trees excel at capturing interactions between variables but struggle with linear effects, whereas logistic regressions exhibit the opposite behavior. Hence, the combination of both allows us to create a more powerful model that also allows for segment-specific insights. More details on the LLM are available in [4].

3.2. Extended hybrid segmented models

Models that can capture non-linearities and interactions well are required to tackle the increasing complexity in data and the growing availability of complex data sources such as texts or images. Often such models are considered as black-box because they are inherently not interpretable.

Hybrid segmented models, however, rely on inherently interpretable models to enable the explainability of the segment-specific insights in the customer churn prediction domain. Despite relative strong performance in benchmarking experiments [4], the inclusion of more powerful models might further enhance the predictive performance as more complex models would in theory allow to capture certain non linear effects and interactions better.

We propose five new hybrid segmented model variants with more powerful models for the segments. Figure 1 visualizes these model variants, for which we consider RF, XGB, ANN, Tabnet and Catboost. Similarly to the inherently interpretable LLM, a decision tree in the first stage segments the data. For each of the segments, a segment-specific model is fitted that allows us to detect and interpret segment-specific drivers. This design reflects that customer segments need to be straightforward to interpret and acceptable to management and based on simple rules [39, 40, 41]. Hence, decision makers can still interpret the different segments, created based on simple but supervised univariate splits. We thus combine interpretable segmentation with black-box predictive capabilities in a single hybrid model.

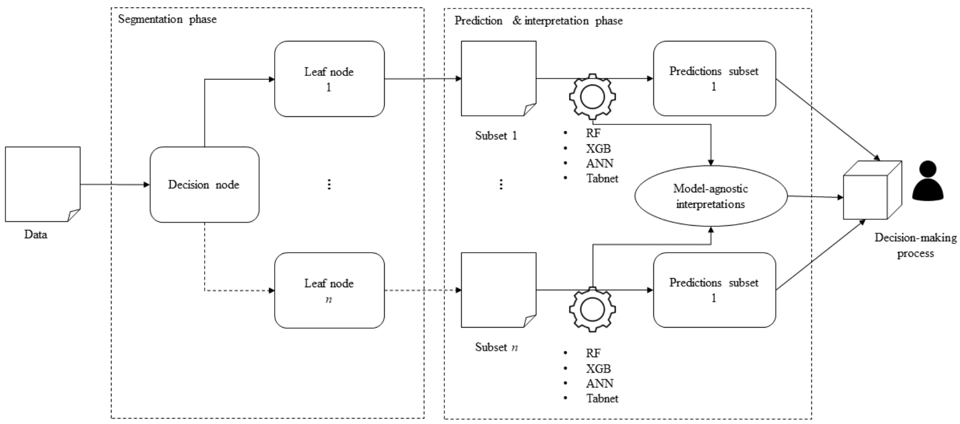


Figure 1: Visual representation of hybrid segmented model for decision making

Furthermore, thanks to the recent advancements in model agnostic methodologies for enhancing model explainability, we are now capable of satisfying decision makers' demands for interpretability, even when employing more advanced and potent models.

3.3. Interpretability

Complex machine learning algorithms can leverage coalition game theory to explain their output. The main idea is to explain how each feature contributed to the final prediction. For this, decision makers often rely on shapley values [e.g. 42, 43] Shapley values present the average marginal contribution of a variable across all possible coalitions, allowing to fairly distribute the contributions to the prediction among all variables [44].

SHAP extends shapley values and enables to explain the output of any machine learning model in an intuitive manner by relying on coalition game theory [9]. SHAP aims to explain the prediction of each customer by computing the contribution of each variable to the prediction of churn, which is the *local accuracy property* of SHAP. As such, it extends Shapley values which assign an importance score to the entire factor and not to each observation [9]. SHAP calculates the importance score for every customer for each variable by considering all subsets of variables without the variable considered and then estimating the effect on the predictions by adding the variable to these subsets [44]. This results in an additive set of marginal contributions for each customer that can be aggregated to yield the prediction for that customer.

There exist several implementations of SHAP. The original implementation described in [9] is an additive feature attribution method, which is a linear model of coalitions [44]. As the main implementation relies on the use of a kernel, it is also known as kernel SHAP. Hence, kernel SHAP allows to estimate SHAP values for any machine learning model by using a specially-weighted local linear regression. As kernel SHAP is relatively slow, it might be impractical to compute for many observations. That is the reason why for tree-based algorithms such as RF or XGB, Lundberg et al. [45] also created treeSHAP which is faster to compute. In addition to being model-specific, treeSHAP can also return unintuitive feature attributions because it changes the value function relying on the conditional expected prediction [44]. As a result of the change in the value function, variables that have no impact on the churn prediction can get a TreeSHAP value different from zero.

To gain insight into specific variables, it is possible to derive feature importance scores from SHAP. Variables with larger absolute Shapley values are more important [44]. Hence, global importance I of the variable j can be calculated as follows [44]:

$$I_j = \frac{1}{n_s} \sum_{i=1}^{n_s} |\alpha_j^{(i)}| \tag{1}$$

,with $\alpha_j^{(i)}$ representing the Shapley value for a given variable j of a customer i.

SHAP allows to derive global interpretability (i.e., across all customers) and local interpretability (i.e., for a given customer), but does not allow for segment specific insights. Therefore, we propose to calculate segment specific importance scores for variables as follows:

$$I_{js} = \frac{1}{n_s} \sum_{i=1}^{n_s} |\alpha_{js}^{(i)}| \tag{2}$$

with $\alpha_{js}^{(i)}$ representing the Shapley value for a given variable j of a customer i within a given segment s.

4. Experimental design

This section introduces data and data preprocessing methods, model evaluation metrics, validation methods, and the hyperparameters considered.

4.1. Data and data preprocessing

The data consists of fourteen customer churn datasets from different sourcers. Table 2 provides an overview of the data sets considered.

Table 2: Overview of data

Data set	Number of customers	Churn rate	Industry group
1	20,000	10.0	Transportation, Communications, Electric, Gas and Sanitary service
2	50,000	7.3	Services
3	32,371	25.2	Retail trade
4	427,833	11.1	Services
5	398,087	4.5	Finance, Insurance and Real Estate
6	602,575	3.2	Finance, Insurance and Real Estate
7	3,827	28.1	Retail trade
8	102,279	6.0	Finance, Insurance and Real Estate
9	47,761	3.7	Services
10	316,578	6.5	Finance, Insurance and Real Estate
11	117,808	3.6	Finance, Insurance and Real Estate
12*	71,047	29.0	Services
13	631,627	2.5	Finance, Insurance and Real Estate
14	573,895	2.6	Finance, Insurance and Real Estate

*Dataset 12 is publicly available [46]

Data sets are pre-processed in a similar manner. First, missing values are imputed using the median for continuous variables and the mode for categorical variables [11]. Next, categorical variables with fewer than 5 categories are transformed using dummy encoding, while for variables with more than 5 categories weight of evidence is used. Third, the values of all variables are rescaled an standardized. Fourth, we apply undersampling in line with other benchmarking studies in the churn prediction domain [4, 14]. Last, we treat outliers using winsorization [4].

4.2. Model Evaluation

The out of sample performance of the benchmark models is compared using the AUC and the TDL [4].

A ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) as the discrimination threshold, or cut-off value chosen to convert posterior probabilities into class membership predictions, is varied. The AUC then summarizes the cut-off independent performance as a single score, and can be interpreted as an indicator of the ranking performance of the model.

From a probabilistic perspective, an AUC of 0.75 reflects a 75% probability that a randomly selected churner is ranked higher than a randomly selected nonchurner. Or formally, $AUC = P(f(x+) > f(x-))$, where $x+$ denotes a randomly selected positive churner and $x-$ denotes a randomly selected negative non-churner. Finally, the AUC is robust to the class imbalance that characterizes most churn datasets [16].

Interpretability is also a crucial characteristic for evaluation measures. For this reason, lift is one of the preferred evaluation metrics for managers [47]. The lift curve depicts how much better a classifier predicts compared to a random selection baseline for different cutoff values. In a business context, not all classifications are equally important. Actions taken based on the model are usually focused on those customers for whom the churn risk is highest. For example, when resources to target customers are limited, organizations want to focus on the specific subset of customers that exhibit the highest risk of churning. As such, the top decile lift better reflects the operational context in which the churn prediction models are applied.

To statistically assess the observed differences, we rely on the Wilcoxon signed rank test to pairwise compare two classifiers over a single or multiple data sets [48]. To compare different classifiers over multiple datasets, we use the Friedman test with Holm post hoc analysis as proposed by [48].

4.3. Validation and Hyper parameters

We use 5x2 cross-validation to evaluate the performance of different classifiers and report results on the test set. Within the training data, an internal cross-validation procedure, based on a single split of 2/3 training and 1/3 validation, selects the optimal hyperparameters for each algorithm. Table 3 provides an overview of the considered hyperparameters for all algorithms. The same hyperparameters are considered for both base models and hybrid models. The hybrid models have the additonal hyperparameters of the decision tree in the first stage as discussed in Section 3. We used a full grid search to optimize hyperparameters, but more efficient ways, such as random searches and/or Baysian optimizaton, could be considered.

Table 3: Overview of considered hyperparameters

Algorithm	Variants[1]	Hyperparameter	Considered values
LLM	H	Threshold pruning	[.01,.10,.15,.20,.25,.30,.35,.40]
	H	No. of observations leaves	N*[.01, .025,.05,.1,.25,.3,.4,.5]
ANN	H	Threshold pruning	[.01,.10,.15,.20,.25,.30,.35,.40]
	H	No. of observations leaves	N*[.01, .025,.05,.1,.25,.3,.4,.5]
	H, B	No. hidden layers	2
	H, B	No. hidden nodes	[64,128]
	H, B	Optimizer	[Adamax, SGD]
	H, B	Dropout	[0.1,0.2]
	H, B	Max. epochs	30
XGB	H	Threshold pruning	[.01,.10,.15,.20,.25,.30,.35,.40]
	H	No. of observations leaves	N*[.01, .025,.05,.1,.25,.3,.4,.5]
	H, B	No. Boosting rounds	[2, 5, 10, 20, 50, 100, 200, 500]
	H, B	Learning rate	[0.001, 0.01, 0.1, 0.2, 0.5]
	H, B	γ	[0.5, 1, 1.5, 2]
RF	H	Threshold pruning	[.01,.10,.15,.20,.25,.30,.35,.40]
	H	No. of observations leaves	N*[.01, .025,.05,.1,.25,.3,.4,.5]
	H, B	No. CART trees	[100, 200, 500, 750, 1000]
	H, B	No. of randomly sampled variables	$\sqrt{(v * [0.1, 0.25, 0.5, 1, 2, 4])}$

[1]H = hybrid, B = Base

5. Results and discussion

In this section, we present and discuss the results obtained from a comprehensive benchmark study, focusing on predictive performance, alongside an insightful case study to explore interpretability.

5.1. Predictive performance

We present and discuss the results of the benchmarking study in two parts. First, we focus on the comparison between the considered base classifiers (i.e., RF, XGB, ANN, Tabnet and Catboost) and their derived hybrid variants as discussed in section 3. Next, we compare the performance of the hybrid models between each other, and with the hybrid LLM, which is established in churn prediction.

5.1.1. Comparison base classifiers with hybrid variants

Results of the base classifiers and their hybrid variants are summarized in Tables A1, A2 and A3 in Appendix A in terms of AUC, TDL and EMPC respectively. The best performing variant is highlighted in bold. Results in Table A1, Table A2 and Table A3 show a similar pattern. Hybrid-XGB tends to perform relatively well in terms of AUC (8 times best algorithm), TDL (5 times best algorithm) and EMPC (7 times best algorithm). In general, hybrid variants perform relatively well. Measured in AUC, a hybrid variant performs best in 10 out of 14 datasets, and on 7 out of 14 datasets for TDL and EMPC.

First, we compare the performance of the base classifiers with their hybrid variant *in all data sets* using a paired Wilcoxon signed rank test. Table 4 summarizes the results of this analysis. These results show that in all data sets there is no significant difference between these hybrid variants and their base classifiers (p-values ≥ 0.10). Therefore, it is possible to generate segment-specific insights without losing predictive performance.

Table 4: Overview significant differences between base classifier and hybrid variant over all datasets

	AUC		*TDL*		*EMPC*	
Comparison	V-statistic	p-value	V-statistic	p-value	V-statistic	p-value
RF vs hybrid RF	71	0.268	78	0.119	46	0.600
XGB vs hybrid XGB	34	0.268	31	0.194	68	0.400
ANN vs hybrid ANN	76	0.153	31	0.194	33	0.400
Tabnet vs hybrid Tabnet	51	0.952	48	0.808	62	0.600
Catboost vs hybrid Catboost	94	0.900	99	0.999	33	0.200

Next, we inspect the results on a more detailed level by comparing the performance of each of the base classifiers with their hybrid variant *in each data set separately.* Such analysis is interesting for decision-makers as they often have to focus on a single case (e.g., predicting churn for one company). Table 5 summarizes these results and indicates what classifier performs significantly better (i.e., base or hybrid) for both AUC and TDL, and all four considered classifiers. A dot (".") indicates non-significant differences between the base and hybrid classifier. This overview shows that for a particular dataset, both the base classier or the hybrid variant could be the preferred variant to optimize predictive performance. For some datasets (e.g., DS 2 or DS 11) the hybrid approach clearly has a beneficial effect on the predictive performance of several classifiers, making it a relevant approach to consider for predicting customer churn.

Table 5: Overview significant differences between base classifier and hybrid variant per dataset

	AUC					*TDL*					*EMPC*				
DS	RF	XGB	ANN	Tabnet	Catboost	RF	XGB	ANN	Tabnet	Catboost	RF	XGB	ANN	Tabnet	Catboost
1	.	base**	hybrid**	.	.	.	.	base**	.	.	.	.	.	.	base**
2	hybrid**	.	.	.	.	.	hybrid**	.	.	hybrid**	.	hybrid**	.	.	.
3	base***	.	.	.	.	.	.	.	.	.	.	.	.	.	.
4	base***	.	.	.	.	base**	.	.	.	base*	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
7	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
8	base***	.	.	.	.	.	.	.	.	.	.	.	.	.	.
9	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
10	base**	hybrid**	.	.	.	.	.	.	.	.	.	.	.	.	.
11	hybrid**	.	.	.	.	hybrid***	hybrid***	.	.	.	.	hybrid**	.	.	.
12	.	.	.	.	.	.	.	.	.	.	.	.	.	.	hybrid**
13	.	.	.	.	.	base***	base*	.	.	.	.	.	.	.	.
14	base***	.	.	hybrid**	.	.	.	.	.	.	.	.	.	.	.

***,**,* indicate significant differences with p-values respectively below 0.01, 0.05 or 0.10

5.1.2. *Comparison between hybrid variants*

In this subsection, we compare the performance of the hybrid variants with the LLM, which has been proposed as an inherently interpretable hybrid model for the prediction of customer churn [4]. Tables A4, Table A5 and Table A6 in Appendix A summarize the results of the LLM and the black-box hybrid models in terms of AUC, TDL, and EMPC respectively. The best performing algorithm is highlighted in bold. Table A4, Table A5 and Table A6 show again a similar pattern. Hybrid XGB performs best on most datasets, for AUC (9 out 14 datasets), TDL (9 out of 14 datasets) and EMPC (8 out of 14 datasets).

First, we compare the hybrid models with the LLM. Table 6 shows the average ranks of the different models over all data sets, which is computed for the Friedman test. Lower ranks indicate better predictive performance. The Friedman test indicates that there are significant differences between the classifiers, so adjusted p-values from the Holm post hoc test are also included. Results show that hybrid XGB performs best (i.e., lowest average rank) for both AUC and TDL, but it is not significantly different from the LLM. Furthermore, the performance of hybrid RF and hybrid Tabnet does not differ significantly from LLM.

The hybrid ANN performs worse than LLM.

Table 6: Average classifier ranks across data sets for AUC, TDL, and EMPC

		AUC	TDL	EMPC
Control:	LLM	2.5	2.28	2.57
Benchmarks:	Hybrid RF	4.86	4.79	3.64
		(0.10)	(0.06)*	(0.13)
	Hybrid XGB	1.43	1.71	.86
		(0.34)	(0.62)	(0.31)
	Hybrid ANN	6.00	5.71	5.93
		(0.00)***	(0.00)***	(0.00)***
	Hybrid Tabnet	2.29	3.21	3.43
		(0.92)	(0.62)	(0.23)
	Hybrid Catboost	3.64	3.14	3.57
		(0.23)	(0.62)	(0.16)

***,**,* indicate significant differences with p-values respectively below 0.01, 0.05 or 0.10

Next, on a more detailed level, we compare the performance of LLM with hybrid XGB for each data set separately analogous to the analysis in Subsection 5.1.1. Table 7 summarizes the significant differences between LLM and hybrid XGB. On individual datasets, hybrid XGB performs significantly better for dataset 13 in terms of AUC and for three different datasets in terms of TDL. LLM never outperforms hybrid XGB. This indicates that on individual datasets, LLM can be outperformed by more complex hybrid models.

Table 7: Overview significant differences between LLM and hybrid XGB per dataset

DS	AUC	TDL	EMPC
1	.	.	Hybrid XGB**
2	.	.	.
3	.	.	.
4	.	.	.
5	.	.	.
6	.	.	.
7	.	Hybrid XGB**	Hybrid XGB**
8	.	.	.
9	.	.	.
10	.	.	.
11	.	Hybrid XGB ***	.
12	.	.	.
13	Hybrid XGB**	Hybrid XGB*	Hybrid XGB*
14	.	.	.

***,**,* indicate significant differences with p-values respectively below 0.01, 0.05 or 0.10

5.2. Case study

In this section, we illustrate our new approach to gain interpretability from the black box hybrid segmented methods on an open data set (i.e., data set 12 in Table 2). We focus on hybrid XGB, as this model has the highest performance (i.e., lowest average rank) over the data sets. All visuals in this case study are based on kernelSHAP.

First, we present the global and segment interpretability of the hybrid XGB model and compare it with the base XGB model. Figure 2 illustrates the hybrid XGB model. It shows that three segments are created based on the *changem_dummy* and *retcalls* variables. These segments can be further analyzed using segment-specific SHAP. The most important variables appear on top for each given segment, based on the Equation 1. Comparing these insights with the base XGB model, which is illustrated in Figure 3, there are some important differences. In the base model, the variable *retcalls* is listed only as the eight most important variable, because it has low impact for most customers, but high impact for some customers. In segment 2, however, this becomes the most important variable. Moreover, relative low feature values tend to have a high impact while this is not the case in other segments or the base model. In the other segments, that variable is of lesser importance. Also, the *recchrge* variable is interesting to discuss. In segment 3, higher values of this variable lead to higher SHAP values, indicating higher importance, while the opposite is true for *recchrge* in the base model. For decision makers, these insights are important as they can steer decision making. For example, for customers in segment three, a high value for *recchrge* might trigger a retention action, while for other customers this is not needed. The segment-specific interpretability of the hybrid XGB also allows one to compare different segments. For example, a certain variable such as *changem_dummy*, is of key importance in segment 3, while in segment 1 it loses all its importance.

Second, the use of more powerful classifiers in the segments allows us to model interaction effects better compared to the LLM. Figure 5 shows a visualization of the LLM based on [4]. It shows that 5 segments are created, which is more than the 3 segments created by the hybrid XGB. Moreover, the segment specific models do not capture any interaction effect. In the proposed visualization of hybrid XGB, however, such insights can be captured. To illustrate this, Figures 4a and Figure 4b provide some examples of non-linear interactions between variables captured by the model in segment 1. Figure 4a shows the interaction between *incalls* and *opeakvce*. Higher values of *incalls* are associated with higher levels of *opeakvce*. We also observe that around the value of 30 for *incalls*, the SHAP value goes to zero while dropping again to -0.15 for higher values. Similarly there is an interaction between *outcalls* and *mou*, which is presented in Figure 4b. Higher values of outcalls are linked with higher levels of mou. The SHAP value of *outcalls* peaks around the value of 75.

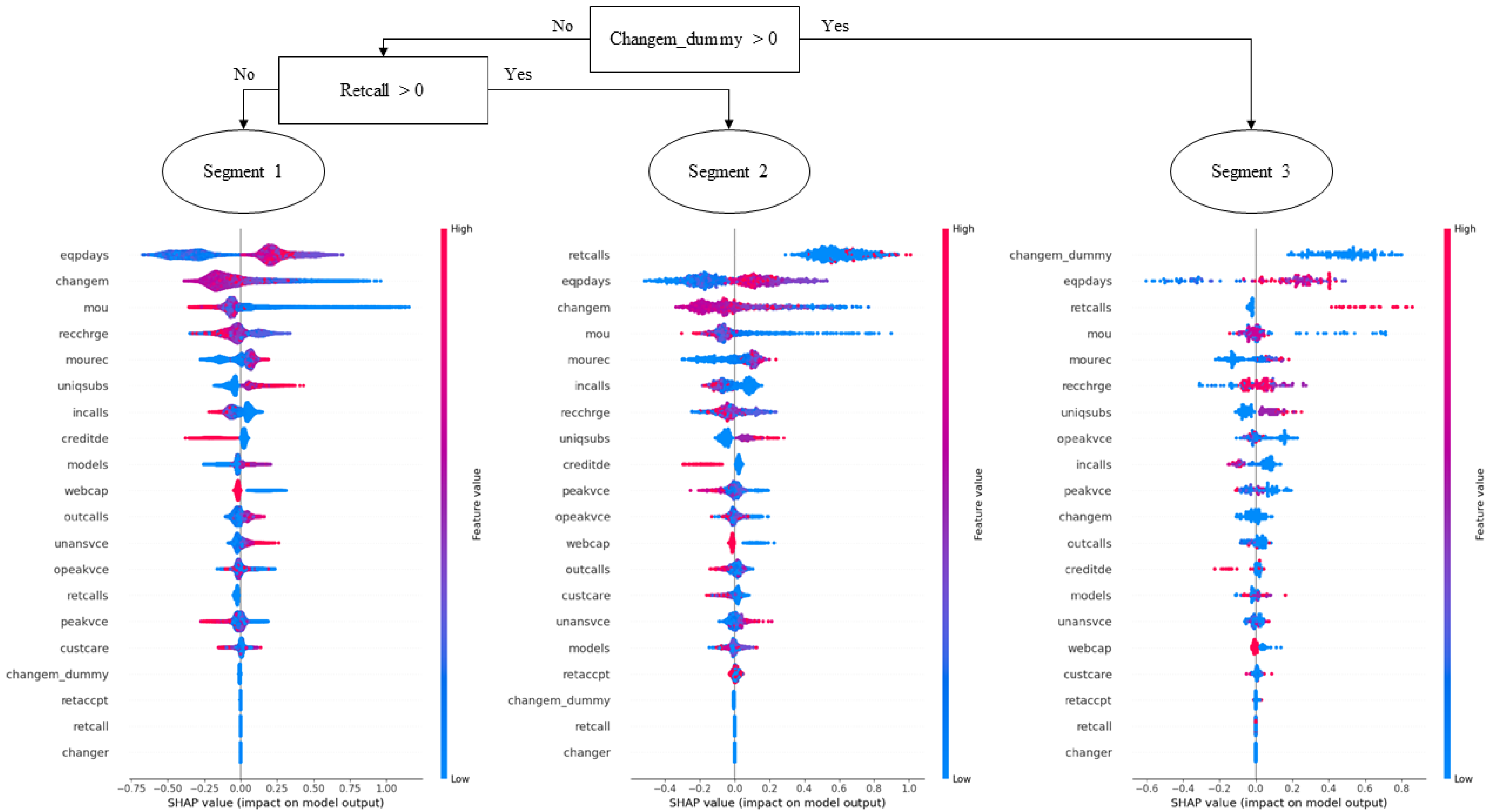


Figure 2: Visual representation of hybrid XGB model

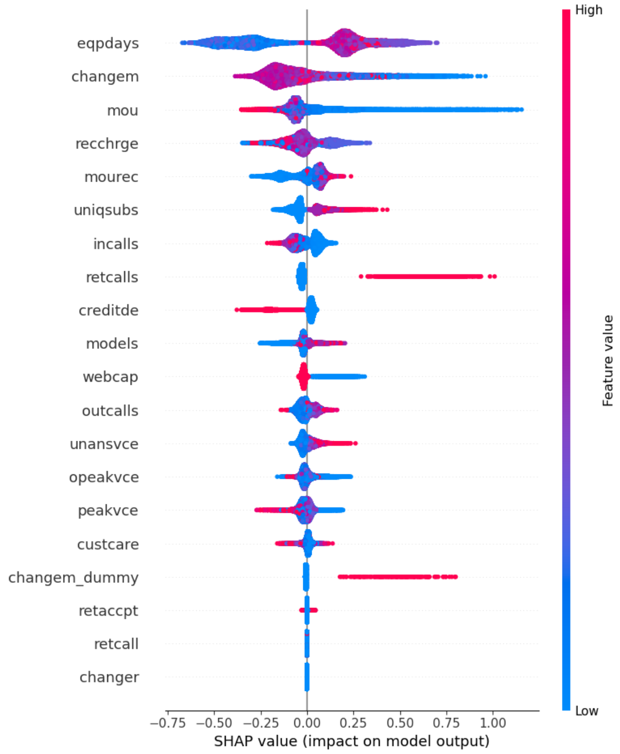
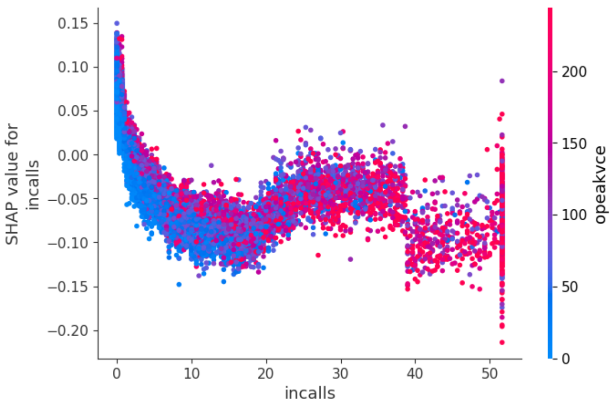
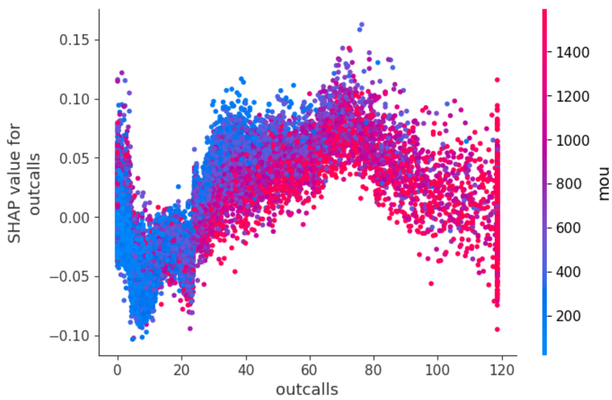


Figure 3: Visual representation of base XGB model using SHAP



(a) Interaction between incalls and opeakvce in segment 1



(b) Interaction between outcalls and mou in segment 1

Figure 4: Examples of segment-specific interaction effects

	1st step: decision tree			2nd step: logistic regression										
Seg.	Rule 1	Rule 2	Rule 3	Shared variables								Segment specific variables		
1	eqpdays ≤ 0.30			incpt. .45	retcalls 1.10	changem -.07	changem_d 0.88	setprccm -.55	creditde -.18			eqpdays .33	directas -.46	
2	eqpdays > 0.30	eqpdays ≤ -0.09		incpt. .19	retcalls .95	changem -.22	changem_d 1.60	setprccm .51	creditde -.44					
3	eqpdays > -0.09	webcap ≤ 0		incpt. .45	retcalls .10				creditde -.27	recharge -.20	custcare -.21			
4	eqpdays > -0.09	webcap > 0	changem ≤ -0.10	incpt. .25	retcalls .15	changem -.30	changem_d 1.22					revenue .28	mou -.23	outcalls -.11
5	eqpdays > -0.09	webcap > 0	changem > 0.10	incpt. .05	retcalls .44				creditde -.19	recharge -.12	custcare -.03	callwait -.32		

Figure 5: Visual representation of logit leaf model

Finally, we can also derive insights on the customer level (i.e., local interpretability). Figure 6 presents the model predictions for a given customer in segment 3 using base XGB and hybrid XGB. In terms of probability, the hybrid model assigns a higher probability (i.e., 0.72 vs. 0.58) which is better as this customer is, in fact, a churner. In the segment specific model, the value of 0 for *retcalls* has a large impact, while this is not the case in the base model. Also the *changem_dummy* variable contributes significantly to the final probability while this is not in the base model. Interestingly, the value of 17.67 of the variable *outcalls* decreases the probability of churn in the base model, while it increases the risk for the segment-specific model. Hence, these segment-specific insights also provide on the customer level important information for decision makers.

Figure 6: Model predictions for customer 1 using base XGB vs. hybrid XGB

6. Conclusions and future research

In this paper we have proposed a novel model-agnostic segmented hybrid-modelling approach that allows for the integration of black-box components into the model, while safeguarding interpretability using SHAP. We show empirically that the resulting approach can achieve superior results and provide novel insights for customer retention management. Our benchmarking results confirm earlier findings of no free lunch in churn prediction. However, compared to LLM, hybrid models can significantly improve predictive performance on some datasets, while also offering increased segment-level interpretability compared to base XGB. Moreover, hybrid segmented modeling can offer unique insights on the customer level, that would not be captured by a non-hybrid variant as we demonstrate in the case study. As such, explainable hybrid segmented models with a non-linear prediction phase offer an exciting paradigm for the data scientist toolbox to boost decision-making for customer retention management. If the analyst can only afford to try one hybrid model, it is advised to use hybrid XGB. In general, hybrid models with tree-based ensemble predictors trump the performance of standard ANNs. While a specialized architecture for tabular data as Tabnet can close the gap with RF, XGB on average still performs best. In light of this observation, we consider tabular deep learning as a first promising avenue for future research in the context of segmented hybrid modelling.

Second, the results of our large benchmarking study show that there exist significant differences between classifiers on specific datasets. Linking of churn dataset characteristics to the optimal method would help with a priori identification of optimal methods. Narrowing down the benchmark selection would save practitioners significant amount of time, compute, and ultimately money. The complexity of datasets might play a role, which might be higher with proprietary datasets compared to open datasets. Future research could look into the impact of dataset complexity on model performance.

Third, the proposed model-agnostic approach to gain insights from hybrid models is an important tool for decision makers, especially in an era that requires responsible analytics. There are significant opportunities in exploring the fairness of segmented models. As companies often both define segment-based as within-segment customer management policies, a separate assessment of the fairness in the segmentation and prediction step may be called for. Therefore, future research might also embed this approach in other settings, such as in applications for healthcare. We relied on a decision tree to assist in segmentation based on actual churn drivers. Alternatively, also, unsupervised methods could be considered in the first stage and should be further explored.

Fourth, the focus of this study lies on the extension and comparison of hybrid models. We relied on previous literature for data preprocessing choices, but future research could future explore the impact of data preprocessing on the performance of (hybrid) models in line with [11].

Fifth, we focus on customer churn prediction in our experiments. Hybrid models could, however, be used in other settings as well, such as credit scoring[49], healthcare[50], or retail [4].

References

[1] G. Torkzadeh, J. C. J. Chang, G. W. Hansen, Identifying issues in customer relationship management at Merck-Medco, Decision Support Systems 42 (2) (2006) 1116–1130. doi:10.1016/j.dss.2005.10.003.

[2] I. Nitzan, B. Libai, Social effects on customer retention, Journal of Marketing 75 (6) (2011) 24–38. doi:10.1509/jm.10.0209.

[3] E. Ascarza, S. A. Neslin, O. Netzer, Z. Anderson, P. S. Fader, S. Gupta, B. G. S. Hardie, A. Lemmens, B. Libai, D. Neal, F. Provost, R. Schrift, In Pursuit of Enhanced Customer Retention Management: Review, Key Issues, and Future Directions, Customer Needs and Solutions (2018). doi:10.1007/s40547-017-0080-0.

[4] A. De Caigny, K. Coussement, K. De Bock, A new hybrid classification algorithm for customer churn prediction based on logistic regression and decision trees, European Journal of Operational Research 269 (2) (2018) 760–772. doi:10.1016/j.ejor.2018.02.009.

[5] T. Verbraken, W. Verbeke, B. Baesens, A novel profit maximizing metric for measuring classification performance of customer churn prediction models, IEEE Transactions on Knowledge and Data Engineering 25 (5) (2013) 961–973.

[6] J. Berrevoets, S. Verboven, W. Verbeke, Treatment effect optimisation in dynamic environments, Journal of Causal Inference 10 (1) (2022) 106–122.

[7] A. De Caigny, K. Coussement, K. W. De Bock, S. Lessmann, Incorporating textual information in customer churn prediction models based on a convolutional neural network, International Journal of Forecasting 36 (4) (2020) 1563–1578.

[8] K. Ljubičić, A. Merćep, Z. Kostanjčar, Churn prediction methods based on mutual customer interdependence, Journal of Computational Science (2023) 101940.

[9] S. M. Lundberg, S. I. Lee, A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems, 2017. arXiv:1705.07874.

[10] S. A. Neslin, S. Gupta, W. Kamakura, L. U. Junxiang, C. H. Mason, Defection detection: Measuring and understanding the predictive accuracy of customer churn models, Journal of Marketing Research 43 (2) (2006) 204–211. doi:10.1509/jmkr.43.2.204.
URL `http://journals.ama.org/doi/abs/10.1509/jmkr.43.2.204`

[11] K. Coussement, S. Lessmann, G. Verstraeten, A comparative analysis of data preparation algorithms for customer churn prediction: A case study in the telecommunication industry, Decision Support

Systems 95 (2017) 27–36. doi:10.1016/j.dss.2016.11.007.
URL http://linkinghub.elsevier.com/retrieve/pii/S0167923616302020

[12] B. Huang, B. Buckley, T. M. Kechadi, Multi-objective feature selection by using NSGA-II for customer churn prediction in telecommunications, Expert Systems with Applications 37 (5) (2010) 3638–3646. doi:10.1016/j.eswa.2009.10.027.
URL http://dx.doi.org/10.1016/j.eswa.2009.10.027

[13] G. Mena, K. Coussement, K. W. De Bock, A. De Caigny, S. Lessmann, Exploiting time-varying RFM measures for customer churn prediction with deep neural networks, Annals of Operations Research (2023). doi:10.1007/s10479-023-05259-9.
URL https://doi.org/10.1007/s10479-023-05259-9

[14] W. Verbeke, K. Dejaeger, D. Martens, J. Hur, B. Baesens, New insights into churn prediction in the telecommunication sector: A profit driven data mining approach, European Journal of Operational Research 218 (1) (2012) 211–229. doi:10.1016/j.ejor.2011.09.031.
URL http://dx.doi.org/10.1016/j.ejor.2011.09.031

[15] E. Stripling, S. vanden Broucke, K. Antonio, B. Baesens, M. Snoeck, Profit maximizing logistic model for customer churn prediction using genetic algorithms, Swarm and Evolutionary Computation 40 (September) (2018) 116–130. doi:10.1016/j.swevo.2017.10.010.
URL http://dx.doi.org/10.1016/j.swevo.2017.10.010 https://doi.org/10.1016/j.swevo.2017.10.010

[16] J. Burez, D. Van den Poel, Handling class imbalance in customer churn prediction, Expert Systems with Applications 36 (3) (2009) 4626–4636.

[17] K. W. De Bock, D. V. D. Poel, An empirical evaluation of rotation-based ensemble classifiers for customer churn prediction, Expert Systems with Applications 38 (10) (2011) 12293–12301. doi:10.1016/j.eswa.2011.04.007.
URL http://dx.doi.org/10.1016/j.eswa.2011.04.007

[18] B. Janssens, M. Bogaert, A. Bagué, D. Van den Poel, B2Boost: instance-dependent profit-driven modelling of B2B churn, Annals of Operations Research (2022). doi:10.1007/s10479-022-04631-5.
URL https://doi.org/10.1007/s10479-022-04631-5

[19] S. Lessmann, J. Haupt, K. Coussement, K. W. De Bock, Targeting customers for profit: An ensemble learning framework to support marketing decision-making, Information Sciences 557 (2021) 286–301. doi:10.1016/j.ins.2019.05.027.

[20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Advances in neural information processing systems 30 (2017).

[21] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, G. Kasneci, Deep neural networks and tabular data: A survey, IEEE Transactions on Neural Networks and Learning Systems (2022).

[22] S. Ö. Arik, T. Pfister, Tabnet: Attentive interpretable tabular learning, in: Proceedings of the Thirty-Fifth AAAI Conference on Artificial Intelligence, 2021, pp. 6679–6687.

[23] K. McDonnell, F. Murphy, B. Sheehan, L. Masello, G. Castignani, Deep learning in insurance: Accuracy and model interpretability using tabnet, Expert Systems with Applications 217 (2023) 119543.

[24] Y. Hwang, J. Song, Recent deep learning methods for tabular data, Communications for Statistical Applications and Methods 30 (2) (2023) 215–226.

[25] I. Bose, X. Chen, Hybrid models using unsupervised clustering for prediction of customer churn, Journal of Organizational Computing and Electronic Commerce 19 (2) (2009) 133–151. doi:10.1080/10919390902821291.

[26] Y. Huang, T. Kechadi, An effective hybrid learning system for telecommunication churn prediction, Expert Systems with Applications 40 (14) (2013) 5635–5647. doi:10.1016/j.eswa.2013.04.020.
URL `http://dx.doi.org/10.1016/j.eswa.2013.04.020`

[27] R. Rajamohamed, J. Manokaran, Improved credit card churn prediction based on rough clustering and supervised learning techniques, Cluster Computing 21 (1) (2018) 65–77. doi:10.1007/s10586-017-0933-1.

[28] E. Sivasankar, J. Vijaya, Hybrid PPFCM-ANN model: an efficient system for customer churn prediction through probabilistic possibilistic fuzzy clustering and artificial neural network, Neural Computing and Applications 31 (11) (2019) 7181–7200. doi:10.1007/s00521-018-3548-4.
URL `https://doi.org/10.1007/s00521-018-3548-4`

[29] A. De Caigny, K. Coussement, W. Verbeke, K. Idbenjra, M. Phan, Uplift modeling and its implications for B2B customer churn prediction: A segmentation-based modeling approach, Industrial Marketing Management 99 (January) (2021) 28–39. doi:10.1016/j.indmarman.2021.10.001.
URL `https://doi.org/10.1016/j.indmarman.2021.10.001`

[30] K. W. De Bock, A. De Caigny, Spline-rule ensemble classifiers with structured sparsity regularization for interpretable customer churn modeling, Decision Support Systems 150 (February) (2021) 113523. doi:10.1016/j.dss.2021.113523.
URL `https://doi.org/10.1016/j.dss.2021.113523`

[31] A. Zadoo, T. Jagtap, N. Khule, A. Kedari, S. Khedkar, A review on churn prediction and customer segmentation using machine learning, in: 2022 International Conference on Machine Learning, Big Data, Cloud and Parallel Computing (COM-IT-CON), Vol. 1, IEEE, 2022, pp. 174–178.

[32] S. P. Adam, S.-A. N. Alexandropoulos, P. M. Pardalos, M. N. Vrahatis, No free lunch theorem: A review, Approximation and Optimization: Algorithms, Complexity and Applications (2019) 57–82.

[33] C. C. Ukwuoma, Z. Qin, M. B. B. Heyat, F. Akhtar, O. Bamisile, A. Y. Muaad, D. Addo, M. A. Al-Antari, A hybrid explainable ensemble transformer encoder for pneumonia identification from chest x-ray images, Journal of Advanced Research 48 (2023) 191–211.

[34] H. Zogan, I. Razzak, X. Wang, S. Jameel, G. Xu, Explainable depression detection with multi-aspect features using a hybrid deep learning model on social media, World Wide Web 25 (1) (2022) 281–304.

[35] W. Verbeke, K. Dejaeger, D. Martens, J. Hur, B. Baesens, New insights into churn prediction in the telecommunication sector: A profit driven data mining approach, European journal of operational research 218 (1) (2012) 211–229.

[36] S. C. Tékouabou, S. C. Gherghina, H. Toulni, P. N. Mata, J. M. Martins, Towards explainable machine learning for bank churn prediction using data balancing and ensemble-based methods, Mathematics 10 (14) (2022) 2379.

[37] C. K. Leung, A. G. Pazdor, J. Souza, Explainable artificial intelligence for data science on customer churn, in: 2021 IEEE 8th International Conference on Data Science and Advanced Analytics (DSAA), IEEE, 2021, pp. 1–10.

[38] H. Guliyev, F. Y. Tatoğlu, Customer churn analysis in banking sector: Evidence from explainable machine learning models, Journal Of Applied Microeconometrics 1 (2) (2021) 85–99.

[39] B. Cooil, L. Aksoy, T. L. Keiningham, Approaches to customer segmentation, Journal of Relationship Marketing 6 (3-4) (2008) 9–39.

[40] A. Hiziroglu, Soft computing applications in customer segmentation: State-of-art review and critique, Expert Systems with Applications 40 (16) (2013) 6491–6507.

[41] C. Rungruang, P. Riyapan, A. Intarasit, K. Chuarkham, J. Muangprathub, Rfm model customer segmentation based on hierarchical approach using fca, Expert Systems with Applications 237 (2024) 121449.

[42] B. Davazdahemami, H. M. Zolbanin, D. Delen, An explanatory machine learning framework for studying pandemics: The case of COVID-19 emergency department readmissions, Decision Support Systems 161 (December 2020) (2022) 113730. doi:10.1016/j.dss.2022.113730.
URL `https://doi.org/10.1016/j.dss.2022.113730`

[43] J. Zhou, C. Wang, F. Ren, G. Chen, Inferring multi-stage risk for online consumer credit services: An integrated scheme using data augmentation and model enhancement, Decision Support Systems 149 (November 2020) (2021) 113611. doi:10.1016/j.dss.2021.113611.
URL `https://doi.org/10.1016/j.dss.2021.113611`

[44] C. Molnar, Interpretable Machine Learning. A Guide for Making Black Box Models Explainable (2nd edition), s.n., 2022.
URL `https://christophm.github.io/interpretable-ml-book`

[45] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, S.-I. Lee, From local explanations to global understanding with explainable ai for trees, Nature Machine Intelligence 2 (1) (2020) 2522–5839.

[46] Kaggle, Datasets for churn telecom, https://www.kaggle.com/datasets/jpacse/datasets-for-churn-telecom (2018).

[47] C. X. Ling, C. Li, Data mining for direct marketing: Problems and solutions., in: Kdd, Vol. 98, 1998, pp. 73–79.

[48] J. Demsar, Statistical Comparisons of Classi ers over Multiple Data Sets, Journal of Machine Learning Research 7 (2006) 1–30.

[49] K. Idbenjra, K. Coussement, A. De Caigny, Investigating the beneficial impact of segmentation-based modelling for credit scoring, Decision Support Systems 179 (2024) 114170. doi:10.1016/j.dss.2024.114170.
URL https://www.sciencedirect.com/science/article/pii/S0167923624000034

[50] M. Kraus, N. Hambauer, K. Müller, P. Kröckel, N. Ulapane, A. De Caigny, K. De Bock, K. Coussement, Coupling Neural Networks Between Clusters for Better Personalized Care, Proceedings of the 57th Hawaii International Conference on System Sciences (2024).
URL https://hdl.handle.net/10125/106821

Appendix A.

Table A1: AUC performance baseline models and hybrid variants

Data Set	RF	Hybrid RF	XGB	Hybrid XGB	ANN	Hybrid ANN	Tabnet	Hybrid Tabnet	Catboost	Hybrid Catboost
1	0.858 (0.011)	0.863 (0.013)	0.865 (0.009)	**0.873** (0.008)	0.808 (0.010)	0.823 (0.011)	0.872 (0.010)	0.866 (0.011)	0.870 (0.009)	0.861 (0.010)
2	0.615 (0.002)	0.619 (0.003)	0.618 (0.002)	**0.621** (0.003)	0.570 (0.003)	0.570 (0.003)	0.620 (0.002)	0.620 (0.003)	0.621 (0.003)	0.619 (0.003)
3	0.815 (0.001)	0.812 (0.001)	0.816 (0.001)	**0.817** (0.001)	0.766 (0.001)	0.766 (0.001)	0.816 (0.001)	0.816 (0.001)	0.809 (0.001)	0.815 (0.001)
4	0.844 (0.001)	0.842 (0.002)	0.848 (0.002)	**0.847** (0.002)	0.797 (0.001)	0.796 (0.002)	0.846 (0.001)	0.846 (0.002)	0.836 (0.002)	0.844 (0.002)
5	**0.797** (0.019)	0.787 (0.011)	0.790 (0.017)	0.789 (0.015)	0.744 (0.016)	0.737 (0.007)	0.793 (0.001)	0.792 (0.013)	0.795 (0.007)	0.791 (0.008)
6	0.864 (0.004)	0.863 (0.007)	0.865 (0.006)	0.867 (0.005)	0.818 (0.004)	0.816 (0.003)	**0.867** (0.002)	0.866 (0.005)	0.845 (0.003)	0.859 (0.004)
7	0.754 (0.009)	0.751 (0.011)	0.754 (0.005)	**0.754** (0.006)	0.704 (0.006)	0.701 (0.007)	0.751 (0.009)	0.754 (0.007)	0.751 (0.003)	0.754 (0.002)
8	0.826 (0.001)	0.823 (0.001)	0.828 (0.001)	**0.828** (0.001)	0.777 (0.001)	0.777 (0.001)	0.827 (0.001)	0.827 (0.001)	0.819 (0.002)	0.827 (0.002)
9	0.724 (0.006)	0.725 (0.005)	**0.728** (0.003)	0.725 (0.005)	0.677 (0.003)	0.676 (0.006)	0.726 (0.005)	0.726 (0.006)	0.727 (0.003)	0.728 (0.004)
10	0.793 (0.002)	0.790 (0.003)	0.794 (0.001)	**0.795** (0.002)	0.744 (0.002)	0.744 (0.002)	0.793 (0.002)	0.793 (0.002)	0.790 (0.003)	0.793 (0.003)
11	0.735 (0.003)	0.739 (0.003)	0.739 (0.005)	**0.740** (0.002)	0.688 (0.002)	0.690 (0.004)	0.737 (0.004)	0.737 (0.003)	0.734 (0.002)	0.736 (0.002)
12	0.624 (0.002)	0.623 (0.002)	0.626 (0.002)	**0.627** (0.003)	0.575 (0.002)	0.577 (0.002)	0.626 (0.002)	0.626 (0.001)	0.621 (0.001)	0.624 (0.001)
13	0.888 (0.003)	0.885 (0.004)	0.886 (0.002)	0.888 (0.002)	0.837 (0.004)	0.834 (0.005)	0.887 (0.004)	**0.889** (0.006)	0.877 (0.003)	0.879 (0.004)
14	0.651 (0.002)	0.646 (0.003)	**0.654** (0.005)	0.653 (0.005)	0.602 (0.005)	0.601 (0.003)	0.648 (0.003)	0.653 (0.005)	0.628 (0.004)	0.643 (0.005)

Table A2: TDL performance baseline models and hybrid variants

Data Set	RF	Hybrid RF	XGB	Hybrid XGB	ANN	Hybrid ANN	Tabnet	Hybrid Tabnet	Catboost	Hybrid Catboost
1	5.294 (0.124)	5.291 (0.083)	5.372 (0.093)	5.359 (0.078)	5.295 (0.090)	5.215 (0.108)	**5.423** (0.119)	5.356 (0.089)	5.383 (0.112)	5.406 (0.099)
2	1.649 (0.169)	1.673 (0.106)	1.681 (0.063)	**1.788** (0.111)	1.613 (0.078)	1.638 (0.116)	1.698 (0.092)	1.751 (0.079)	1.687 (0.689)	1.742 (0.705)
3	2.734 (0.033)	2.714 (0.023)	2.775 (0.070)	2.789 (0.042)	2.703 (0.048)	2.683 (0.050)	2.790 (0.031)	2.777 (0.062)	**2.792** (0.042)	2.780 (0.055)
4	**4.454** (0.028)	4.422 (0.052)	4.502 (0.029)	4.514 (0.036)	4.401 (0.031)	4.412 (0.041)	4.486 (0.033)	4.492 (0.037)	4.500 (0.050)	4.482 (0.047)
5	3.873 (0.207)	3.721 (0.268)	3.787 (0.242)	**3.895** (0.358)	3.688 (0.203)	3.767 (0.225)	3.725 (0.223)	3.834 (0.121)	3.710 (0.242)	3.820 (0.198)
6	5.306 (0.072)	5.293 (0.065)	5.345 (0.075)	5.384 (0.062)	5.199 (0.110)	5.250 (0.067)	5.318 (0.110)	5.324 (0.087)	**5.475** (0.085)	5.364 (0.105)
7	2.083 (0.099)	2.075 (0.142)	2.196 (0.098)	**2.232** (0.148)	1.991 (0.116)	2.063 (0.124)	2.082 (0.120)	2.166 (0.117)	2.075 (0.086)	2.102 (0.098)
8	4.476 (0.094)	4.418 (0.066)	4.522 (0.059)	4.535 (0.078)	4.395 (0.048)	4.405 (0.030)	4.536 (0.099)	4.483 (0.068)	4.545 (0.082)	**4.545** (0.082)
9	3.338 (0.366)	3.296 (0.193)	3.168 (0.349)	3.261 (0.296)	3.205 (0.248)	3.263 (0.317)	3.206 (0.475)	**3.413** (0.258)	3.256 (0.400)	3.263 (0.546)
10	3.721 (0.028)	3.714 (0.049)	3.790 (0.075)	**3.807** (0.043)	3.695 (0.042)	3.669 (0.056)	3.789 (0.067)	3.770 (0.053)	3.804 (0.028)	3.806 (0.017)
11	2.353 (0.026)	2.396 (0.032)	2.423 (0.024)	**2.469** (0.032)	2.322 (0.030)	2.331 (0.022)	2.431 (0.023)	2.436 (0.023)	2.415 (0.036)	2.401 (0.039)
12	1.516 (0.090)	1.541 (0.093)	**1.619** (0.068)	1.616 (0.098)	1.504 (0.077)	1.498 (0.094)	1.617 (0.080)	1.589 (0.099)	1.616 (0.092)	1.576 (0.084)
13	5.417 (0.033)	5.378 (0.033)	**5.506** (0.018)	5.488 (0.024)	5.373 (0.027)	5.366 (0.027)	5.476 (0.027)	5.477 (0.026)	5.450 (0.045)	5.467 (0.062)
14	2.100 (0.244)	2.097 (0.338)	**2.241** (0.222)	2.109 (0.372)	1.988 (0.340)	2.033 (0.135)	2.013 (0.218)	1.959 (0.253)	2.103 (0.182)	2.054 (0.204)

Table A3: EMPC performance baseline models and hybrid variants

Data Set	RF	Hybrid RF	XGB	Hybrid XGB	ANN	Hybrid ANN	Tabnet	Hybrid Tabnet	Catboost	Hybrid Catboost
1	0.871 (0.013)	0.875 (0.022)	**0.892** (0.041)	0.891 (0.005)	0.854 (0.024)	0.865 (0.041)	0.884 (0.025)	0.867 (0.002)	0.886 (0.032)	0.876 (0.022)
2	0.023 (0.008)	0.022 (0.005)	0.021 (0.009)	**0.026** (0.014)	0.010 (0.003)	0.010 (0.017)	0.011 (0.009)	0.014 (0.004)	0.016 (0.006)	0.017 (0.018)
3	8.839 (0.035)	8.820 (0.035)	8.835 (0.077)	8.829 (0.018)	8.800 (0.032)	8.770 (0.049)	**8.846** (0.024)	8.808 (0.074)	8.814 (0.032)	8.812 (0.044)
4	2.793 (0.002)	2.765 (0.001)	**2.795** (0.015)	2.783 (0.01)	2.746 (0.008)	2.754 (0.009)	2.768 (0.001)	2.766 (0.013)	2.764 (0.005)	2.765 (0.001)
5	**0.465** (0.014)	0.462 (0.011)	0.459 (0.003)	0.460 (0.007)	0.449 (0.007)	0.453 (0.022)	0.459 (0.011)	0.461 (0.009)	0.462 (0.009)	0.462 (0.019)
6	0.870 (0.030)	0.866 (0.002)	0.858 (0.020)	0.870 (0.034)	0.864 (0.026)	0.861 (0.042)	0.875 (0.042)	**0.883** (0.034)	0.872 (0.025)	0.872 (0.011)
7	9.255 (0.170)	9.667 (0.124)	9.548 (0.010)	**9.676** (0.151)	9.216 (0.110)	9.449 (0.011)	9.631 (0.084)	9.621 (0.063)	9.640 (0.135)	9.657 (0.022)
8	2.505 (0.013)	2.765 (0.017)	2.771 (0.011)	2.767 (0.014)	2.481 (0.004)	2.473 (0.005)	**2.791** (0.025)	2.774 (0.024)	2.765 (0.015)	2.770 (0.005)
9	0.033 (0.011)	0.037 (0.017)	0.041 (0.003)	0.040 (0.007)	0.041 (0.014)	0.036 (0.012)	0.033 (0.013)	0.037 (0.017)	**0.044** (0.002)	0.036 (0.005)
10	0.190 (0.011)	0.190 (0.005)	0.191 (0.008)	**0.192** (0.006)	0.159 (0.005)	0.180 (0.008)	0.190 (0.004)	0.190 (0.009)	0.181 (0.002)	0.187 (0.001)
11	0.018 (0.021)	0.015 (0.017)	0.021 (0.002)	**0.022** (0.006)	0.020 (0.005)	0.011 (0.022)	0.022 (0.008)	0.021 (0.004)	0.018 (0.012)	0.013 (0.009)
12	8.666 (0.030)	8.656 (0.016)	**8.676** (0.011)	8.656 (0.020)	8.612 (0.017)	8.644 (0.004)	8.655 (0.016)	8.651 (0.019)	8.638 (0.021)	8.659 (0.001)
13	2.729 (0.026)	2.765 (0.031)	**2.797** (0.032)	2.774 (0.020)	2.752 (0.024)	2.764 (0.027)	2.755 (0.025)	2.770 (0.022)	2.768 (0.026)	2.774 (0.018)
14	0.241 (0.016)	0.254 (0.076)	**0.274** (0.062)	0.258 (0.071)	0.240 (0.061)	0.231 (0.020)	0.252 (0.061)	0.254 (0.02)	0.235 (0.022)	0.245 (0.038)

Table A4: AUC results LLM and blackbox hybrid variants

Data Set	LLM	Hybrid RF	Hybrid XGB	Hybrid ANN	Hybrid Tabnet	Hybrid catboost
1	0.869 (0.013)	0.863 (0.013)	**0.873** (0.008)	0.823 (0.011)	0.866 (0.011)	0.861 (0.010)
2	0.619 (0.003)	0.619 (0.003)	**0.621** (0.003)	0.570 (0.003)	0.620 (0.003)	0.619 (0.003)
3	0.816 (0.001)	0.812 (0.001)	**0.817** (0.001)	0.766 (0.001)	0.816 (0.001)	0.815 (0.001)
4	**0.847** (0.002)	0.842 (0.002)	0.847 (0.002)	0.796 (0.002)	0.846 (0.002)	0.844 (0.002)
5	**0.793** (0.012)	0.787 (0.011)	0.789 (0.015)	0.737 (0.007)	0.792 (0.013)	0.791 (0.008)
6	0.866 (0.006)	0.863 (0.007)	**0.867** (0.005)	0.816 (0.003)	0.866 (0.005)	0.859 (0.004)
7	0.753 (0.006)	0.751 (0.011)	0.754 (0.006)	0.701 (0.007)	**0.754** (0.007)	0.754 (0.002)
8	0.827 (0.002)	0.823 (0.001)	**0.828** (0.001)	0.777 (0.001)	0.827 (0.001)	0.827 (0.002)
9	0.724 (0.003)	0.725 (0.005)	0.725 (0.005)	0.676 (0.006)	0.726 (0.006)	**0.728** (0.004)
10	0.795 (0.002)	0.790 (0.003)	**0.795** (0.002)	0.744 (0.002)	0.793 (0.002)	0.793 (0.003)
11	0.739 (0.004)	0.739 (0.003)	**0.740** (0.002)	0.690 (0.004)	0.737 (0.003)	0.736 (0.002)
12	0.626 (0.002)	0.623 (0.002)	**0.627** (0.003)	0.577 (0.002)	0.626 (0.001)	0.624 (0.001)
13	0.884 (0.005)	0.885 (0.004)	0.888 (0.002)	0.834 (0.005)	**0.889** (0.006)	0.879 (0.004)
14	0.649 (0.004)	0.646 (0.003)	**0.653** (0.005)	0.601 (0.003)	0.653 (0.005)	0.643 (0.005)

Table A5: TDL results LLM and blackbox hybrid variants

Data Set	LLM	Hybrid RF	Hybrid XGB	Hybrid ANN	Hybrid Tabnet	Hybrid Catboost
1	5.366 (0.083)	5.291 (0.083)	5.359 (0.078)	5.215 (0.108)	5.356 (0.089)	**5.406** (0.099)
2	1.765 (0.107)	1.673 (0.106)	**1.788** (0.111)	1.638 (0.116)	1.751 (0.079)	1.742 (0.705)
3	2.784 (0.071)	2.714 (0.023)	**2.789** (0.042)	2.683 (0.050)	2.777 (0.062)	2.780 (0.055)
4	4.509 (0.043)	4.422 (0.052)	**4.514** (0.036)	4.412 (0.041)	4.492 (0.037)	4.482 (0.047)
5	3.866 (0.255)	3.721 (0.268)	**3.895** (0.358)	3.767 (0.225)	3.834 (0.121)	3.820 (0.198)
6	5.364 (0.082)	5.293 (0.065)	**5.384** (0.062)	5.250 (0.067)	5.324 (0.087)	5.364 (0.105)
7	2.085 (0.067)	2.075 (0.142)	**2.232** (0.148)	2.063 (0.124)	2.166 (0.117)	2.102 (0.098)
8	4.494 (0.234)	4.418 (0.066)	4.535 (0.078)	4.405 (0.030)	4.483 (0.068)	**4.545** (0.082)
9	3.340 (0.085)	3.296 (0.193)	3.261 (0.296)	3.263 (0.317)	**3.413** (0.258)	3.263 (0.546)
10	3.795 (0.028)	3.714 (0.049)	**3.807** (0.043)	3.669 (0.056)	3.770 (0.053)	3.806 (0.017)
11	2.408 (0.033)	2.396 (0.032)	**2.469** (0.032)	2.331 (0.022)	2.436 (0.023)	2.401 (0.039)
12	**1.618** (0.057)	1.541 (0.093)	1.616 (0.098)	1.498 (0.094)	1.589 (0.099)	1.576 (0.084)
13	5.472 (0.039)	5.378 (0.033)	**5.488** (0.024)	5.366 (0.027)	5.477 (0.026)	5.467 (0.062)
14	**2.198** (0.319)	2.097 (0.338)	2.109 (0.372)	2.033 (0.135)	1.959 (0.253)	2.054 (0.204)

Table A6: EMPC results LLM and blackbox hybrid variants

Data Set	LLM	Hybrid RF	Hybrid XGB	Hybrid ANN	Hybrid Tabnet	Hybrid Catboost
1	0.877 (0.002)	0.875 (0.022)	**0.891** (0.005)	0.865 (0.041)	0.867 (0.002)	0.876 (0.022)
2	0.023 (0.019)	0.022 (0.005)	**0.026** (0.014)	0.010 (0.017)	0.014 (0.004)	0.017 (0.018)
3	8.825 (0.037)	8.820 (0.035)	**8.829** (0.018)	8.770 (0.049)	8.808 (0.074)	8.812 (0.044)
4	2.767 (0.012)	2.765 (0.001)	**2.783** (0.010)	2.754 (0.009)	2.766 (0.013)	2.765 (0.001)
5	**0.463** (0.001)	0.462 (0.011)	0.460 (0.007)	0.453 (0.022)	0.461 (0.009)	0.462 (0.019)
6	**0.877** (0.039)	0.866 (0.002)	0.870 (0.034)	0.861 (0.042)	0.883 (0.034)	0.872 (0.011)
7	9.632 (0.036)	9.667 (0.124)	**9.676** (0.151)	9.449 (0.011)	9.621 (0.063)	9.657 (0.022)
8	2.767 (0.007)	2.765 (0.017)	2.767 (0.014)	2.473 (0.005)	**2.774** (0.024)	2.770 (0.005)
9	**0.041** (0.008)	0.037 (0.017)	0.040 (0.007)	0.036 (0.012)	0.037 (0.017)	0.036 (0.005)
10	0.190 (0.006)	0.190 (0.005)	**0.192** (0.006)	0.180 (0.008)	0.190 (0.009)	0.187 (0.001)
11	0.018 (0.002)	0.015 (0.017)	**0.022** (0.006)	0.011 (0.022)	0.021 (0.004)	0.013 (0.009)
12	8.659 (0.01)	8.656 (0.016)	8.656 (0.02)	8.644 (0.004)	8.651 (0.019)	**8.659** (0.001)
13	2.767 (0.015)	2.765 (0.031)	2.774 (0.02)	2.764 (0.027)	2.770 (0.022)	**2.774** (0.018)
14	0.248 (0.061)	0.254 (0.076)	**0.258** (0.071)	0.231 (0.02)	0.254 (0.02)	0.245 (0.038)